

Lecture Notes 2

Economic Growth and the Solow Model

Intermediate Macroeconomics, EC2211

Martin Flodén

Fall 2025

Department of Economics, Stockholm University

Contents and literature

- The production function
- Growth accounting
- The Solow model

Literature:

- Jones (2024) chapters 3-5

Production

The production function

The Cobb Douglas production function

$$Y = F(K, L) = AK^\alpha L^{1-\alpha}$$

Y = output, GDP

$F(.,.)$ = production function

A = total factor productivity (TFP)

K = capital input

L = labor input

α = capital share in production, $\alpha \in (0, 1)$

Math repetition: derivatives

Let $y = f(x)$

The derivative of y with respect to x is written

$$\frac{dy}{dx} = \frac{df(x)}{dx} = f'(x)$$

Math repetition: derivatives

Let $y = f(x_1, x_2)$

The derivatives of y with respect to its arguments, x_1 and x_2 , are written:

$$\frac{\partial y}{\partial x_1} = \frac{\partial f(x_1, x_2)}{\partial x_1} = f'_{x_1}(x_1, x_2)$$

and

$$\frac{\partial y}{\partial x_2} = \frac{\partial f(x_1, x_2)}{\partial x_2} = f'_{x_2}(x_1, x_2)$$

Math repetition: derivatives

Recall that if $y = f(x) = x^\alpha$, then $f'(x) = \alpha x^{\alpha-1}$

Examples:

- If $f(x) = x + 0.5x^2 - x^4$, then $f'(x) = 1 + x - 4x^3$
- If $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, then $f'_{x_1}(x_1, x_2) = \alpha x_1^{\alpha-1} x_2^{1-\alpha}$

The production function

- How much does production increase if we increase capital but hold labor constant?

The marginal product of capital, MPK :

$$MPK = \frac{\partial Y}{\partial K} = \frac{\partial F(K, L)}{\partial K} = F_K$$

- How much does production increase if we increase labor but hold capital constant?

The marginal product of labor, MPL :

$$MPL = \frac{\partial Y}{\partial L} = \frac{\partial F(K, L)}{\partial L} = F_L$$

Constant returns to scale

Consider again the Cobb-Douglas production function:

$$Y = AK^{\alpha}L^{1-\alpha}$$

If all production factors are increased by a factor x , production also increases by factor x :

$$F(xK, xL) = A(xK)^{\alpha}(xL)^{1-\alpha} = Ax^{(\alpha+1-\alpha)}K^{\alpha}L^{1-\alpha} = xF(K, L)$$

- If, for example, $F(K, L) = 10$ then $F(3K, 3L) = 30$
- A function with this property is said to display **constant returns to scale**

Diminishing marginal products

We still consider the Cobb-Douglas production function $Y = AK^\alpha L^{1-\alpha}$

The marginal products of capital and labor are

$$MPK = \alpha AK^{\alpha-1} L^{1-\alpha} = \alpha A \left(\frac{L}{K} \right)^{1-\alpha},$$

and

$$MPL = (1 - \alpha) AK^\alpha L^{-\alpha} = (1 - \alpha) A \left(\frac{K}{L} \right)^\alpha$$

- If we increase K but hold L constant, production will increase but less and less as K becomes larger
- If we increase L but hold K constant, production will increase but less and less as L becomes larger
- The production function displays **diminishing marginal products**

Diminishing marginal product of capital

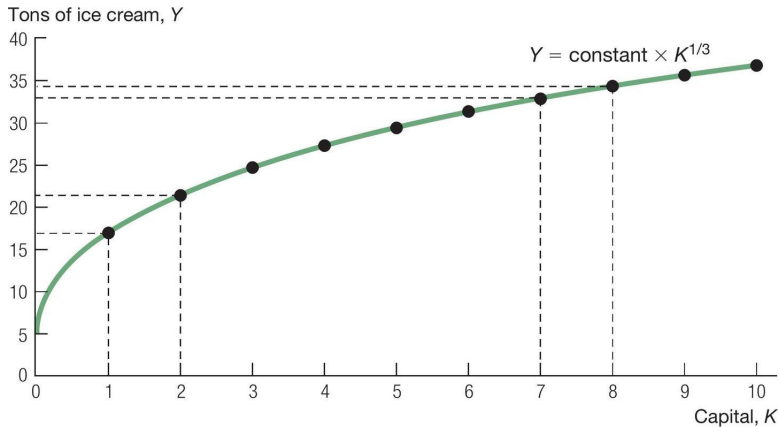
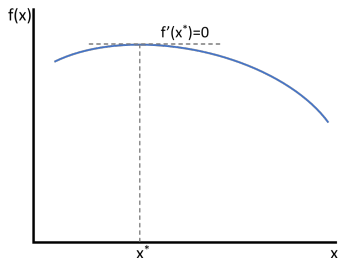


Figure 4.1 in Jones (2024)

Math repetition: profit maximization

- Consider a function $f(x)$
- Recall that $f'(x)$ is the slope of f in point x
- If some x^* is maximizes the function, then $f'(x^*) = 0$
- This is the **first-order condition** (f.o.c.)
- (But note that the same holds for a minimum; the f.o.c. is a *necessary* but not sufficient condition)



Profit maximization under Cobb-Douglas

Competitive firms choose capital and labor to maximize profits:

$$\max_{K,L} \Pi(K, L) = \underbrace{AK^\alpha L^{1-\alpha}}_{F(K,L)} - rK - wL$$

where Π denotes the profit function, w the wage and r the rental rate of capital

First-order conditions for profit maximization:

$$\frac{\partial \Pi}{\partial K} = \underbrace{\alpha AK^{\alpha-1} L^{1-\alpha}}_{MPK} - r = 0 \quad (1)$$

$$\frac{\partial \Pi}{\partial L} = \underbrace{(1-\alpha)AK^\alpha L^{-\alpha}}_{MPL} - w = 0 \quad (2)$$

The demand for capital and labor

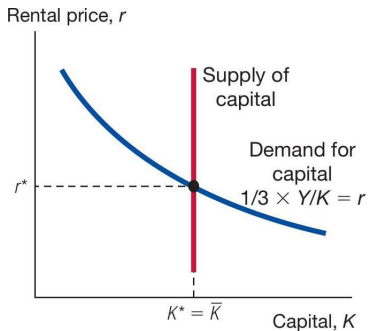
Re-arranging (1):

$$MPK = \alpha AK^{\alpha} L^{(1-\alpha)} K^{-1} = \alpha \frac{Y}{K} = r \quad (3)$$

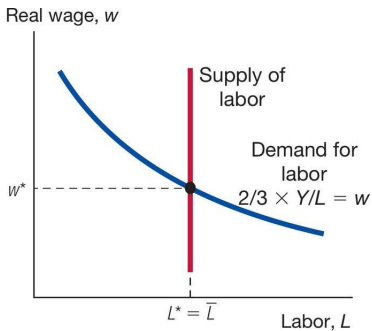
Re-arranging (2):

$$MPL = (1 - \alpha) AK^{\alpha} L^{(1-\alpha)} L^{-1} = (1 - \alpha) \frac{Y}{L} = w \quad (4)$$

Equilibrium in the capital and labor markets



(a) The capital market



(b) The labor market

Figure 4.2 in Jones (2024)

Note that (3) and (4) imply:

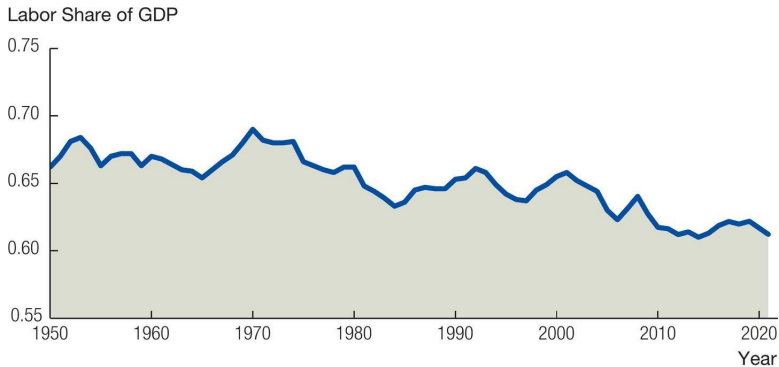
$$\alpha = \frac{rK}{Y}$$

and

$$1 - \alpha = \frac{wL}{Y}$$

Jones (2024) assumes that $\alpha = 1/3$. Why?

Labor's share of GDP ($1 - \alpha$)



Source: U.S. Department of Commerce and author's calculations, Bureau of Economic Analysis, Table 1.10, [bea.gov](https://www.bea.gov).

Figure 2.3 in Jones

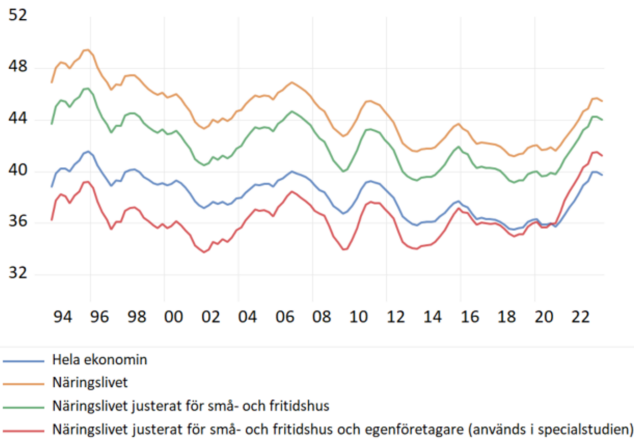
Labor's share of GDP

- Macroeconomists often say that the labor's share of GDP in the United States has been roughly constant at around $2/3$ for the last century
- Data for the U.S. and many other countries however indicate that the labor's share has declined somewhat since the 1980s (see previous slide and [this note](#))
 - The reasons behind this fall are not obvious
 - Separating compensation to labor from compensation to capital owners is not always trivial
- Sweden may be an exception (has no clear trend)
 - See next slide and [this blog post](#)
(The slide and the blog post are in Swedish. Some translations: "vinstandelar" = "profit shares", "hela ekonomin" = "total economy", "näringslivet" = "private sector")

Capital's share of GDP, Sweden (α)

Diagram 1 Vinstandelar

Procent



Anm. Fyra kvartals glidande medelvärde

Källor: SCB och Konjunkturinstitutet.

Growth accounting

Growth accounting: Can we quantify the contribution to growth from different factors?

- Consider our Cobb-Douglas production function:

$$Y_t = A_t K_t^\alpha L_t^{1-\alpha}$$

- Apply the rules for growth rates:

$$g_Y = g_A + \alpha g_K + (1 - \alpha)g_L$$

- This shows that the growth rate of output is the sum of TFP growth and (weighted) growth of the production factors
- We can observe or estimate Y_t , K_t , L_t and α in the data
- Growth in TFP is then calculated as the residual

$$g_A = g_Y - \alpha g_K - (1 - \alpha)g_L$$

Cross-country comparisons require expressions in per-capita terms

In per capita terms, production is

$$y_t = A_t(K_t/L_t)^\alpha = A_t k_t^\alpha$$

so:

$$g_y = g_A + \alpha g_k$$

where $y \equiv Y/L$ and $k \equiv K/L$ denote GDP and capital per person ("per capita"), respectively

Income per capita

Can we use the production function to "explain" why some countries are so much richer than others?

Cross-country comparisons require expressions in per-capita terms

Recall the production function $Y = AK^\alpha L^{1-\alpha}$ and divide by L :

$$y \equiv \frac{Y}{L} = \frac{AK^\alpha L^{1-\alpha}}{L} = AK^\alpha L^{-\alpha} = Ak^\alpha,$$

where $y = Y/L$ and $k \equiv K/L$ denote GDP and capital per person ("per capita"), respectively

Predicted per capita GDP

- Suppose $y = Ak^{1/3}$ and that A is the same as in the United States in all countries
- Actual differences in production are much larger than predicted by differences in capital input (see next two slides!)
- If this is the production function, much of differences in GDP per capita must be attributed to TFP

Predicted per capita GDP relative to the U.S. if $y = k^{1/3}$

With no difference in productivity, the model predicts smaller differences in income across countries than we observe.

TABLE 4.3

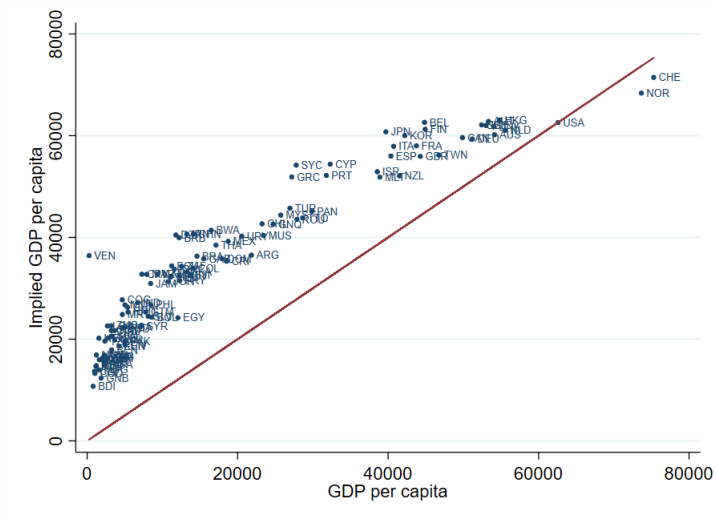
The Model's Prediction for Per Capita GDP (U.S. = 1)

Country	Observed capital per person, $\bar{k}$	Predicted per capita GDP $y = \bar{k}^{1/3}$	Observed per capita GDP
United States	1.000	1.000	1.000
Switzerland	1.488	1.142	1.203
Japan	0.914	0.971	0.634
Italy	0.792	0.925	0.651
Spain	0.716	0.895	0.645
U.K.	0.714	0.894	0.707
Brazil	0.195	0.580	0.233
China	0.275	0.650	0.226
South Africa	0.165	0.548	0.200
India	0.082	0.434	0.107
Burundi	0.005	0.172	0.013

Predicted per capita GDP is computed as $\bar{k}^{1/3}$, that is, assuming no differences in productivity across countries. Data correspond to the year 2019 and are divided by the values for the United States. Source: Penn World Table, Version 10.0.

- Consider, for example, China: the table shows that $k^{China}/k^{USA} = 0.275$
- If $y = k^{1/3}$, then $y^{China}/y^{USA} = (k^{China}/k^{USA})^{1/3} = 0.275^{1/3} = 0.650$
- Actual GDP per capita in China is just 22.6 percent of GDP per capita in the United States

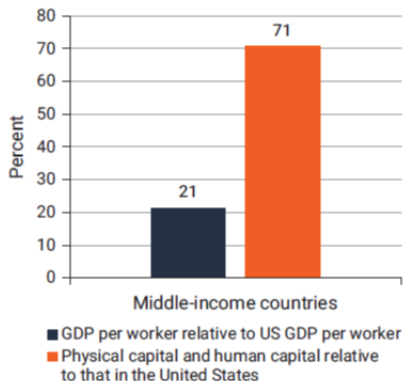
Predicted per capita GDP



This graph shows similar information as the table on the previous page but for more countries. Horizontal axis: Actual GDP per capita in 1919 (2017 USD) from PWT 10.01. Vertical axis: Predicted GDP per capita according to $Ak^{1/3}$ where A is based on the U.S. value and k calculated from PWT with the same method as Jones. Ireland and Qatar were dropped from sample.

Much more differences in GDP per capita than in capital per capita

Figure O.2 If capital accumulation were enough, work in middle-income countries would be nearly three-quarters as rewarding as in the United States, not just one-fifth



If capital is the only production factor, TFP in most other countries must be much lower than in the U.S. to match the data

TABLE 4.4

Measuring TFP So the Model Fits Exactly

Country	Per capita GDP (y)	$\bar{k}^{1/3}$	Implied TFP ($\bar{A}$)
United States	1.000	1.000	1.000
Switzerland	1.203	1.142	1.054
U.K.	0.707	0.894	0.792
Japan	0.634	0.971	0.654
Italy	0.651	0.925	0.703
Spain	0.645	0.895	0.721
Brazil	0.233	0.580	0.402
South Africa	0.200	0.548	0.365
China	0.226	0.650	0.347
India	0.107	0.434	0.247
Burundi	0.013	0.172	0.074

Calculations are based on the equation $y = \bar{A}\bar{k}^{1/3}$. Implied productivity $\bar{A}$ is calculated from data on y and $\bar{k}$ for the year 2019, so that this equation holds exactly as $\bar{A} = y/\bar{k}^{1/3}$.

In order for our model to match the data, poor countries must be very inefficient in production; that is, they must have low TFP.

Table 4.4 in Jones

Why does TFP differ across countries?

- The accounting exercise in Jones (2024) suggests that A is twice as important as k in explaining cross-country differences in y
- Why does TFP differ across countries?
 - (Labor input)
 - Human capital
 - Technology
 - Institutions
 - Misallocation

Another indication of the importance of TFP

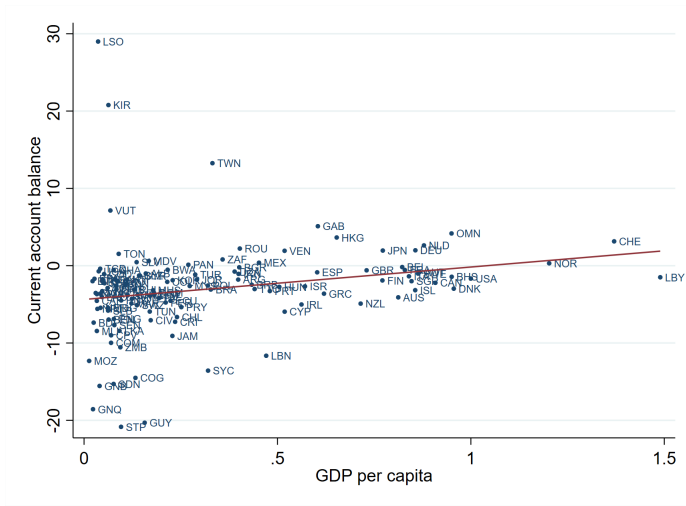
- Suppose that all countries have identical production functions $Y = AK^\alpha L^{1-\alpha}$ and identical A
- Suppose that physical capital can move freely between countries
- Then capital should flow from rich to poor countries because the MPK would then be much higher in poor countries (Robert Lucas, 1990, "Why doesn't capital flow from rich to poor countries")
- The current account balance shows capital flows: $CA = S - I$
 - A negative current account balance means that capital flows to the country (investment is larger than saving)
 - There is some tendency for capital to flow to poor countries
 - But not at all to the extent this "model" predicts

Some tendency for capital to flow from rich to poor countries in 2010s



Averages for 2010-2019. Current account balance in percent of GDP. GDP per capita relative to USA. Source: IMF WEO.

Modest capital flows in 1980s



Averages for 1980-1989. Current account balance in percent of GDP. GDP per capita relative to USA. Source: IMF WEO.

The Solow model

The Solow model

- The production function is the key ingredient
- Does capital accumulation generate growth?
- A key model for
 - ... understanding economic growth and its limitations
 - ... but also a building stone for much other macroeconomic analysis

The Solow model

- Solow (1956), "A Contribution to the Theory of Economic Growth"
- Nobel prize 1987: "The study of the factors which permit production growth and increased welfare has been a central feature in economic research for many years. Robert M. Solow's prize recognizes his exceptional contributions in this area."



Robert Solow 1924-2023

Notation

Y_t : output at time t

A : total factor productivity (TFP)

K_t : the capital stock at time t

L_t : labor input at time t

I_t : investment at time t

C_t : consumption at time t

K_0 : initial stock of capital

δ : the depreciation rate of capital¹

s : savings rate

¹ Jones uses $\bar{\delta}$ to denote the depreciation rate, but δ is a much more common notation

Production

Production is a function of the two production factors, capital and labor:

$$Y_t = F(K_t, L_t) \quad (5)$$

We assume that the production function exhibits *constant returns to scale* and *diminishing marginal products of capital and labor*. Closed-economy resource constraint:

$$C_t + I_t = Y_t \quad (6)$$

Capital accumulation

Capital in period $t + 1$ is determined by:

$$K_{t+1} = K_t + I_t - \delta K_t \quad (7)$$

Equation (7) can be re-arranged to read:

$$\Delta K_{t+1} = \underbrace{I_t - \delta K_t}_{\text{net investment}} \quad (8)$$

where $\Delta K_{t+1} \equiv K_{t+1} - K_t$

In each period $t > 0$, K_t results from previous investment and depreciation

To get the model started, an initial value for K_0 is needed

Consumption and investment

Assume that a constant fraction of income is saved (and invested):

$$I_t = sY_t \quad (9)$$

This implies that consumption is:

$$C_t = (1 - s)Y_t \quad (10)$$

Solving the model

Assume for now that there is no population growth so that labor input is also constant over time, $L_t = L$, and we also assume that TFP is constant over time, $A_t = A$

Equations (8), (9) and (5) imply:

$$\Delta K_{t+1} = sF(K_t, L) - \delta K_t \quad (11)$$

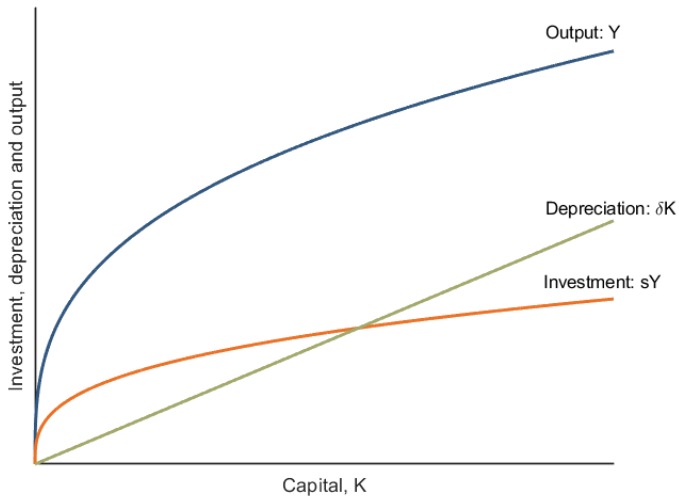
Cobb-Douglas production function

Assume that the production function is Cobb-Douglas:

$$Y_t = AK_t^\alpha L^{1-\alpha} \quad (12)$$

- Recall that the marginal product of capital is then diminishing:
 - We have assumed that labor input is fixed
 - Plotted against capital, Y and sY will have the concave shape we saw previously
- Depreciation, δK , is however *linear*
- The change in the capital stock from one period to the next is investment minus depreciation, i.e. the difference between sY and δK

The Solow Diagram



Steady state

Steady state

In a steady state, all variables are constant

If capital is constant, then $K_{t+1} = K_t$, i.e. $\Delta K_{t+1} = 0$

Equation (11) implies that the steady-state level of capital K^* must satisfy:

$$sY^* = sF(K^*, L) = \delta K^* \quad (13)$$

where Y^* is steady-state output

Steady state under Cobb-Douglas

We assume that the production function is Cobb-Douglas,

$$Y_t = AK_t^\alpha L^{1-\alpha} \quad (14)$$

Equation (13), $sY^* = \delta K^*$, then implies that

$$sA(K^*)^\alpha L^{1-\alpha} = \delta K^*$$

Steady state under Cobb-Douglas

Solving for K^* :

$$K^* = \left(\frac{sA}{\delta} \right)^{\frac{1}{1-\alpha}} L \quad (15)$$

Plugging (15) into (14) yields steady-state output:

$$Y^* = A^{\frac{1}{1-\alpha}} \left(\frac{s}{\delta} \right)^{\frac{\alpha}{1-\alpha}} L \quad (16)$$

Steady-state capital and output per worker

Dividing (15) by L we obtain steady-state capital per worker:

$$k^* \equiv \frac{K^*}{L} = \left(\frac{sA}{\delta} \right)^{\frac{1}{1-\alpha}} \quad (17)$$

Dividing (16) by L we obtain steady-state output per worker:

$$y^* \equiv \frac{Y^*}{L} = A^{\frac{1}{1-\alpha}} \left(\frac{s}{\delta} \right)^{\frac{\alpha}{1-\alpha}} \quad (18)$$

The Solow Diagram

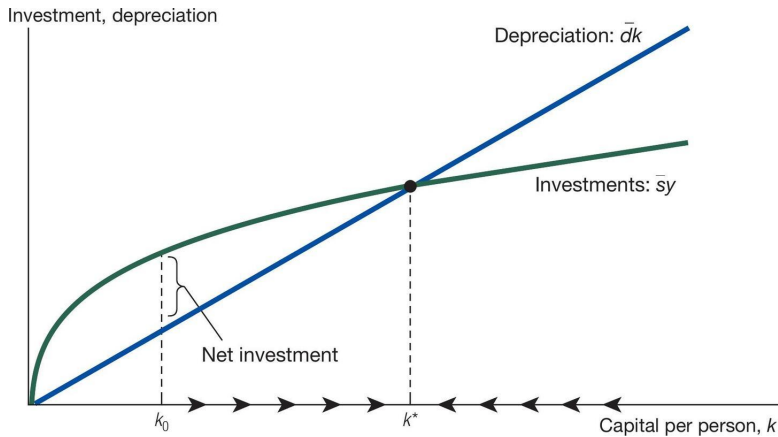


Figure 5.1 in Jones (2024)

The Solow Diagram with output

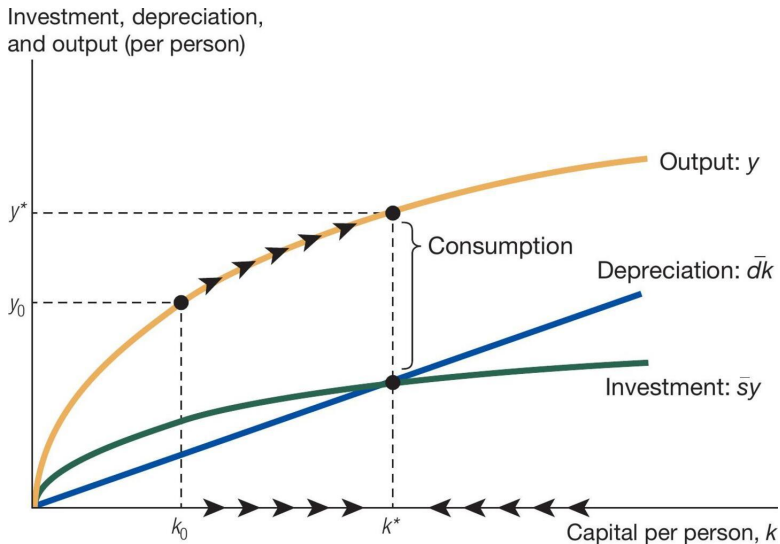
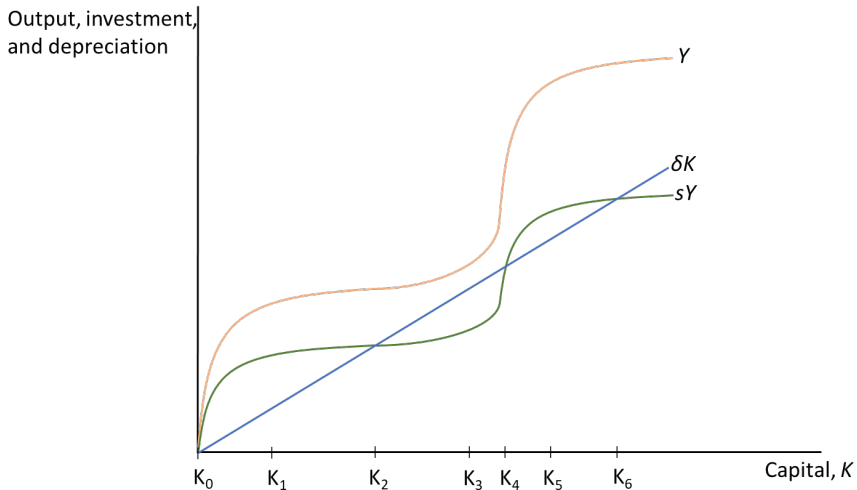


Figure 5.2 in Jones (2024).

Exercise: Where is the steady state? (Poverty trap?)

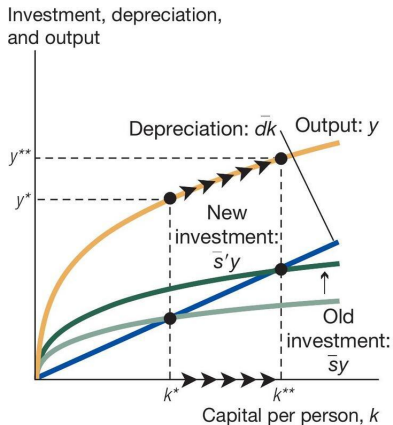


Note that this (non-standard) production function is not Cobb Douglas

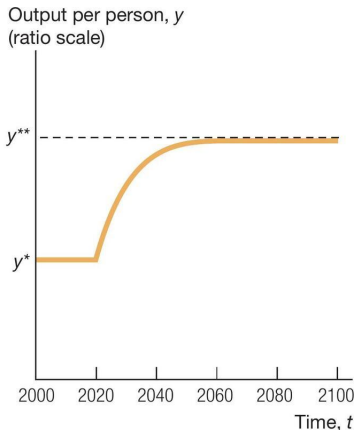
Some model implications

- More capital accumulation (higher s) leads to higher income but not to sustained growth
- Why does the economy reach a steady state?
 - Diminishing returns to capital, but depreciation is proportional to capital
 - In the steady state, investment is just offset by capital depreciation
- The capital-output ratio in steady state is proportional to the savings rate
- The savings rate can be inefficiently high ("Golden Rule", see exercise)

An increase in the investment rate



(a) The Solow diagram with output.



(b) Output per person over time.

Figure 5.5 in Jones (2024)

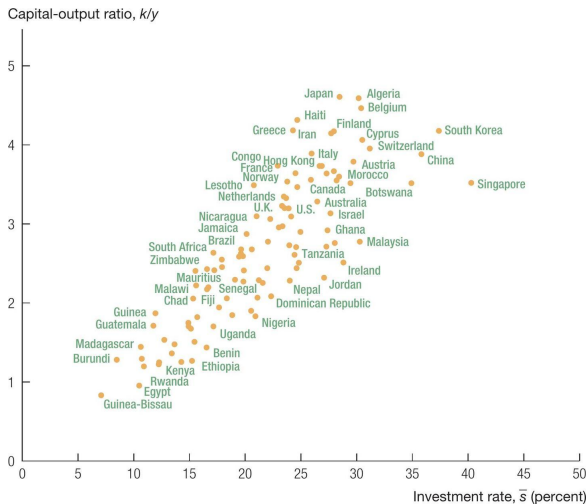
Capital-output ratio and savings

Recall that equation (13) predicts $sY^* = \delta K^*$, so that:

$$\frac{K^*}{Y^*} = \frac{s}{\delta} \quad (19)$$

Idea: plot capital-income ratios against investment rates

"Explaining" capital in the Solow model



Source: Penn World Table, Version 10.0. The capital-output ratio is measured in the year 2019, while the investment rate is averaged over the period 1990 to 2019.

Figure 5.3 in Jones (2024)

Capital-output ratio and savings

- Figure 5.3 looks like very strong support for the model in the data
- But how do we measure capital stocks?
- Typically measured indirectly from investment rates
- Jones assumes that $\delta = 0.05$ and calculates
 - $K_{1960} = K^*$
 - $K_{1961} = (1 - \delta)K_{1960} + I_{1960}$, etc

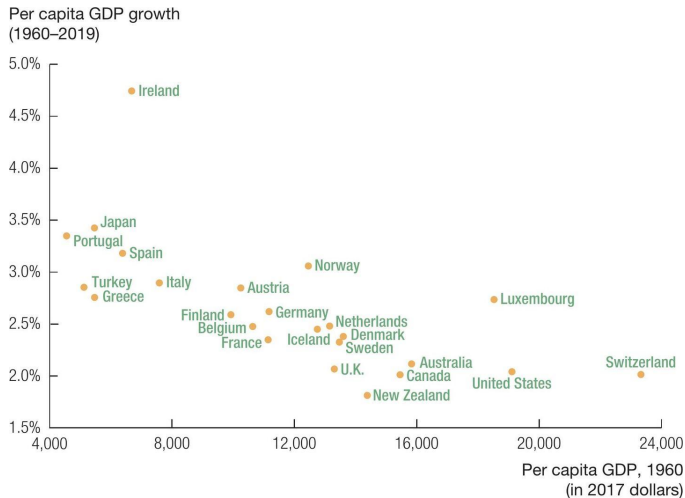
Transition dynamics

- No long-run growth in this version of the Solow model: the economy settles down at K^* and Y^*
- But countries can grow temporarily during transitions to steady states
 - If $K_t < K^*$ the growth rate will be positive
 - If $K_t > K^*$ the growth rate will be negative
 - The further from the steady state, the larger the (absolute value of) growth rates
- The growth that we observe may be due to shocks that cause deviations from steady states, or to shifts in steady states

Transitional dynamics, convergence

- Suppose that countries have similar production functions & similar TFP
 - They then have similar steady states
 - But maybe some (the "poor") just have little capital
 - Poor countries are then far from steady state, should grow rapidly
- What does the data say?

Growth rates in the OECD, 1960-2019



Source: Penn World Table, Version 10.0.

Figure 5.8 in Jones (2024)

Growth rates around the world, 1960-2019

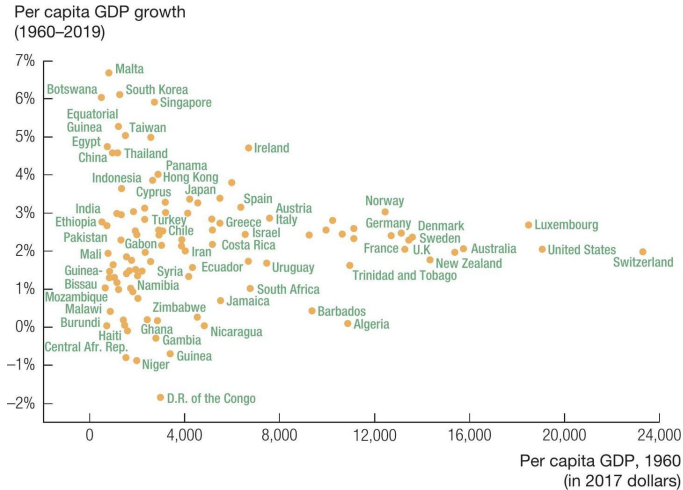


Figure 5.9 in Jones (2024)

Convergence

- Clear evidence of convergence among OECD countries since 1960
- Less clear evidence of convergence in larger sample of countries
- Possible interpretation:
 - OECD countries are similar and have similar steady-states
 - But this is not the case for other economies; they may be close to their steady-states
- Convergence vs. conditional convergence

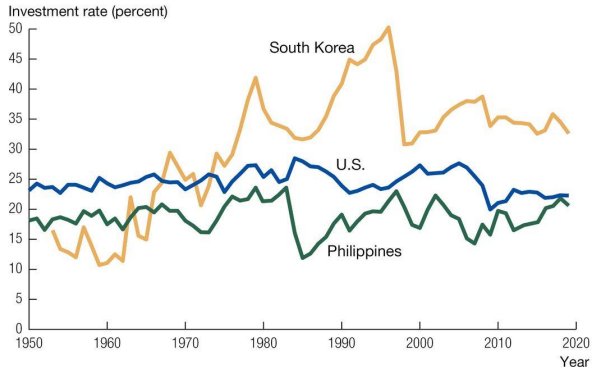
South Korea and the Philippines 1960-2019

	GDP per capita relative to USA 1960	GDP per capita relative to USA 2019	Increase in GDP per capita 1960-2019 (percent)
USA	1.00	1.00	228
Philippines	0.11	0.13	317
Korea	0.07	0.67	3261

Can the Solow model explain why growth was so much more rapid in Korea than in the Philippines?

Case study: South Korea and the Philippines 1960-2019

The investment rate in South Korea rose sharply between 1960 and 1995, while it remained relatively stable in the United States and the Philippines.



Source: Penn World Table, Version 10.0.

Figure 5.10 in Jones (2024)

Summing up

- We have analyzed the basic Solow model with constant L and A
- The model shows that capital accumulation:
 - can generate transitional growth
 - cannot generate sustained growth
 - can be inefficiently high (exercise)
- Taking the model to the data, much cross-country variation remains unexplained
- Next time: population growth, productivity, human capital, natural resources, ...